Northeastern University
Institute for Intelligent Networked Systems

Day 2 Intro

Welcome!

Outline

- Agenda
- Hackathon scoring
- Q&A

Agenda for today

Time	Session – EXP 610
9:00–9:30	**Q&A on Day 1** Open floor for issues, blockers, or questions.
9:30–11:15	**Training 3: Contributing to OpenAirInterface (Enrichment)** Deep dive into how you can contribute to OpenAirInterface. First steps, understanding the codebase, tests, review process. Discuss good first contributions and small items on the roadmap.
11:15–11:30	**Break**
11:30–13:00	**Training 4: Introduction to dApps (Enrichment)** What are dApps? Primer on the concept, framework, and role in the O-RAN ecosystem. Conceptual overview for participants new to the topic.
13:00–14:00	**Lunch**
14:00–18:00+	**Hackathon – Session 2** Core development time. Tutors available (at least one per team). Teams should be actively implementing their scheduling algorithm and beginning integration testing.

Practicalities

- 6 teams
  - Coordinate with your team!
- There will be one tutor per team
  - Ask questions if you get stuck, if something doesn't work, etc
- We will review progress at the end of each day

Day 2 Milestones

- Scheduling algorithm implementation substantially underway
- Major technical blockers identified and addressed

Hackathon Scoring

Goals - recap:

- Know more about OpenAirInterface and RAN software
- Get a hands-on experience on infrastructure already set up for you - focus on software and functionalities
- Understand how to contribute to open-source software for 5G systems
- Understand the role of this software in your research

- No scripted numerical scoring
  - This gives more flexibility to explore different approaches, develop parts you think are interesting, etc
- Performance + approach + presentation

Hackathon Scoring

Prepare two slides

Slide 1 — What You Did & Results

- Approach: describe your scheduling algorithm and design choices
- Results: provide 2/3 plots
  - Throughput per UE
  - Latency: measured end-to-end delay distribution for ping
    - Highlight latency violations

Slide 2 — Open-Source Software & Research

- Why does open-source software matter for wireless research?
- How did tools like OAI, OpenRAN Gym, or X5G enable your work?
- What would have been harder — or impossible — without open platforms?
- Any contributions or improvements you'd consider feeding back upstream?

Hackathon Scoring

- **Deadline for slides: Wednesday 10pm**
  - Results should be ready by Wednesday ~5pm, then take care of slides

One person per group to present

Thursday Workshop

	• Panelists: Howard Thomas (VIAVI), Tingjun Chen (Duke University), Richard Rouil (NIST), Vijay Shah (North Carolina State University)
3:00–3:15	**Break**
3:15–3:45	**Hackathon Review and Presentations**
3:45–4:30	**Panel 3: Open-Source Development and Testing: The Challenges with RAN Software** • Moderator: Michele Polese (Northeastern) • Panelists: Lincoln Lavoie (UNH IOL), Robert Schmidt (OpenAirInterface), Kirk Webb (University of Utah/POWDER), Peter Pacheco (MITRE), Damien Saucez (INRIA/SLICES)

Q&A

- Any questions?
- Any issues/blockers from yesterday?